

Addis Ababa University Collage of Health Sciences School of medicine

Biostatistics Work sheet II

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1. It has been estimated that about 30% of frozen chicken contain enough salmonella bacteria to cause illness if improperly cooked. A consumer purchases 12 frozen chickens. What is the probability that the consumer will have more than 6 contaminated chickens?
0.079
2. The probability that a certain machine will produce a defective item is 0.20. If a random sample of 6 items is taken from the output of this machine, what is the probability that there will be 5 or more defectives in the sample?**0.0016**
3. Fifteen percent of new residential central air conditioning units installed by a supplier need additional adjustments requiring a service call. Assume that a recent sample of seven such units constitutes a Bernoulli process. Interest centers on X, the number of units among these seven that need additional adjustments. The mean and variance of X are, respectively
4. Consider 8 blood donors chosen randomly from a population. The probability that the donor has type A blood is .40. Which of the following is CORRECT?
 - A. The probability of 1 or fewer donors having type A blood is about .11.
 - B. The probability of 7 or more donors NOT having type A blood is about .0087.
 - C. The probability of exactly 5 donors having type A blood is about .28.
 - D. The probability of exactly 5 donors NOT having type A blood is about .12.
 - E. The probability that between 3 and 5 donors (inclusive) will have type A blood is about .37.
5. The time required to assemble a microscope is normally distributed with a mean of 12 minutes and a standard deviation of 1.5 min. Find the probability that a particular assembly takes more than 14.25 minutes
6. If the total cholesterol values for a certain target population are approximately normally distributed with a mean of 200 (mg/100 mL) and a standard deviation of 20 (mg/100 mL), the probability that a person picked at random from this population will have a cholesterol value greater than 240 (mg/100 mL) is _____
 - i. What value of Cholesterol cuts off the upper 2.5 % of this population? _____
 - ii. What value of cholesterol cuts off the lower 5% of the population _____

7. The following table lists the probability distribution of the number of hematology test performed in a day at Zewditu hospital

X	$p(x=x)$
0	0.05
1	0.25
2	0.30
3	0.22
4	0.18

- i. What is the expected number of test performed with in a day? **2.23**
 - ii. What is the standard deviations? **2.03**
8. Suppose that in a certain malarious area past experience indicate that the probability of a person with a high fever will be positive for malaria is 0.7. Consider 3 randomly selected patient (with high fever) in the same area.
- i. What is the probability that no patient will be positive for malaria?
 - ii. What is the probability that exactly one of the patient's will be positive for malaria?
9. The following is a contingency table, classifying 1000 employees of a company according to their gender and this promotion status in the past year

	Promoted (A)	Not promoted (B)	Total
Male (M)	250	350	600
Female (F)	<u>125</u>	<u>275</u>	<u>400</u>
	375	625	1000

- A. What is the probability that an employee taken at random would be a male and was promoted? **0.25**
- B. What will be the probability that an employee be promoted given that gender status is female? **0.125**
- C. Are events "gender" and "promotion" dependent events? **yes**

10. Given a binomial variable with the mean of 20 and variance of 16. Find n and P

11. Suppose the average length of stay in a chronic disease hospital of a certain type of patient is 60 days with standard deviations of 15. If it is reasonable to assume an approximately normal distribution of stay. Find the probability that a randomly selected patient from this group will have a length of stay;

- I. Greater than 50 day
- II. Between 30 and 60 days
- III. Less than 30 days
- IV. Greater than 90 days

10. The probability that a person suffering from migraine headache will obtain relief with a particular drug is 0.9. Three randomly selected sufferers from migraine are given the drug. Find the probability that the number obtaining relief will be :

- i. Exactly zero
- ii. Exactly one
- iii. More than one
- iv. Two or fewer

11. Given a normally distributed population with a mean of 75 and a variance of 625. Find

- i. $P(50 \leq x \leq 100)$
- ii. $P(x > 90)$
- iii. $P(x \geq 85)$

12. Given the following probability. Find C

- i. $P(Z \leq C) = 0.0055$
- ii. $P(-C < Z < C) = 0.8132$
- iii. $P(C < Z < 2.98) = 0.1117$

